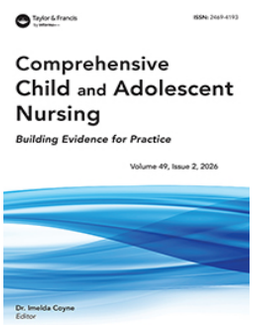
Taylor & Francis
Taylor & Francis Group



Comprehensive Child and Adolescent Nursing

ISSN: 2469-4193 (Print) 2469-4207 (Online) Journal homepage: www.tandfonline.com/journals/icpn21

Do Children and Adolescents with Attention Deficit Hyperactivity Disorder Use Screens More? A Systematic Review and Meta-Analysis

Paulo Gonçalves Júnior, Italo Emmanoel Silva e Silva, Marcela Baraúna Magno, Lucianne Cople Maia, Matheus Melo Pithon & Raildo da Silva Coqueiro

To cite this article: Paulo Gonçalves Júnior, Italo Emmanoel Silva e Silva, Marcela Baraúna Magno, Lucianne Cople Maia, Matheus Melo Pithon & Raildo da Silva Coqueiro (2026) Do Children and Adolescents with Attention Deficit Hyperactivity Disorder Use Screens More? A Systematic Review and Meta-Analysis, Comprehensive Child and Adolescent Nursing, 49:2, 95-118, DOI: 10.1080/24694193.2026.2640837

To link to this article: https://doi.org/10.1080/24694193.2026.2640837

View supplementary material

Published online: 26 Mar 2026.

Submit your article to this journal

Article views: 191

View related articles

View Crossmark data

Citing articles: 1 View citing articles

Full Terms & Conditions of access and use can be found at
https://www.tandfonline.com/action/journalInformation?journalCode=icpn21

COMPREHENSIVE CHILD AND ADOLESCENT NURSING
2026, VOL. 49, NO. 2, 95–118
https://doi.org/10.1080/24694193.2026.2640837

Taylor & Francis
Taylor & Francis Group

Check for updates

Do Children and Adolescents with Attention Deficit Hyperactivity Disorder Use Screens More? A Systematic Review and Meta-Analysis

Paulo Gonçalves Júnior[a], Italo Emmanoel Silva e Silva[b], Marcela Baraúna Magno[c,d], Lucianne Cople Maia[c], Matheus Melo Pithon[c], and Raildo da Silva Coqueiro[a]

[a]Post-Graduate Program in Physical Education, Universidade Estadual do Sudoeste da Bahia (UESB), Jequié, Brazil; [b]Post-Graduate Program in Nursing and Health, UESB, Jequié, Brazil; [c]Department of Pediatric Dentistry and Orthodontics, School of Dentistry, Universidade Federal do Rio de Janeiro, Rio de Janeiro, Brazil; [d]Post-Graduation Program in Dentistry, Universidade Veiga de Almeida, Rio de Janeiro, Brazil

ABSTRACT

Background: Our aim in conducting this review was to investigate whether children and adolescents with attention deficit hyperactivity disorder (ADHD) exhibit differences in screen time when compared to children and adolescents with typical development.

Method: The study protocol was registered in the PROSPERO database. Databases and gray literature were screened. Studies that compared screen time between children and adolescents with ADHD and those with typical development were included and critically evaluated with the Joanna Briggs Institute tools. Meta-analyses were performed considering screen time ≥2 hours/day or its mean (hours/day). The Grading of Recommendations Assessment, Development, and Evaluation (GRADE) approach was applied for evaluating the certainty of the evidence. Twenty-two studies were included.

Results: In total, 235 283 children and adolescents of both sexes were evaluated. ADHD was diagnosed clinically or based on caregiver reporting, and different screen types were reported. With a low and very low certainty of evidence, children and adolescents with ADHD were more likely to spend ≥2 hours/day during the week with computers and screens in general and had a higher mean screen time ($p < .05$). ADHD medication did not seem to influence these results.

Conclusions: Children and adolescents with ADHD spent more time during weekdays using screens and were therefore more likely to exhibit excessive screen and computer use.

ARTICLE HISTORY
Received 25 February 2025
Accepted 27 February 2026

KEYWORDS
ADHD; children; adolescents; sedentary behavior; screen time; typical development

Introduction

Attention deficit hyperactivity disorder (ADHD) is a neurodevelopmental condition that typically emerges in childhood and persist into adulthood (American Psychiatric Association, 2014). Its presentation may vary significantly across individuals and contexts, and diagnostic criteria continue to evolve. Additionally, the manifestation of symptoms can differ notably between males and females, with hyperactivity

CONTACT Raildo da Silva Coqueiro rscoqueiro@uesb.edu.br Department of Health 1, Universidade Estadual do Sudoeste da Bahia, Rua José Moreira Sobrinho, S/N, Jequiezinho, Jequié, Bahia 45210-506, Brazil.
Supplemental data for this article can be accessed online at https://doi.org/10.1080/24694193.2026.2640837
© 2026 Taylor & Francis Group, LLC

sometimes presenting as cognitive restlessness rather than overt physical behavior (Musálek et al., 2020). While challenges related to attention and impulsivity are often highlighted, it is important to acknowledge that individuals with ADHD may also exhibit strengths such as creativity, hyperfocus, and innovative thinking (American Psychiatric Association, 2014).

ADHD is associated with developmental differences that may impact personal, social, academic, and/or occupational functioning (Arango-Tobón et al., 2023; Ayano et al., 2023; Dona et al., 2023). However, estimating its true prevalence remains challenging due to variability in diagnostic criteria and methodologies across studies. Global prevalence estimates in children and adolescents range from 3,4% a 13,9%, with a predominance in males (Ayano et al., 2023). This gender difference may, in part, be attributed to diagnostic biases and the underrecognition of symptoms in females, who often present with less overt forms of inattention or internalizing behaviors. Moreover, diagnostic delays are common, particularly when symptoms are mild, masked by coping strategies, or occur in environments that do not pose significant demands on attention and behavioral regulation (Ayano et al., 2023).

In recent decades, rapid changes in humanity's daily habits, related to environmental, physical, economic, and social factors, have occurred (Fujiwara et al., 2022). One issue that has drawn attention and concerned health agencies is the increase in time spent in sedentary behavior among school-aged youth (Mahumud et al., 2021). Such behavior is recognized as a major risk factor for various adverse physical and mental health outcomes in children and adolescents, such as obesity, sleep disturbances, anxiety, and lower academic performance (Saunders & Vallance, 2017). Sedentary behavior is also associated with morbidity and mortality in adults and older adults (Katzmarzyk et al., 2019).

In children and adolescents, most of the time spent in sedentary behavior involves screen use as part of activities such as watching television, playing video games, and using a computer, tablet, or smartphone, among others (Saunders & Vallance, 2017). Over the last two decades, screen time has increased substantially in this population, due to greater access to mobile technologies, streaming services, and digital platforms in daily life (Saunders & Vallance, 2017). Given this context, most of the literature on sedentary behavior in children and adolescents is based on the assessment of screen time (Kim et al., 2011).

Some evidence suggests that screen time has a greater negative impact on individuals with ADHD than on people with typical development. These impacts may involve both physical and mental health domains, such as increased risk of sleep problems, anxiety symptoms, and difficulties with self-regulation (Merrill et al., 2021; Verma et al., 2022). However, many studies have focused on the potential adverse effects rather than on the amount of time spent on screens (Merrill et al., 2021; Verma et al., 2022). There is still no consensus in the literature on whether screen time duration differs between children and adolescents with ADHD and those with typical development (Ebenegger et al., 2012; Párraga et al., 2019). Therefore, our aim in undertaking this systematic review was to answer the following question: do children and adolescents with ADHD exhibit differences in screen time when compared to children and adolescents with typical development?

Materials and methods

Protocol and registration

The study protocol was registered in the PROSPERO database under the number CRD42022327389 and is accessible at the following direct link: https://www.crd.york.ac.uk/PROSPERO/view/CRD42022327389.

Following protocol registration, minor methodological adjustments were required based on the characteristics of the eligible studies. The original protocol was designed to evaluate sedentary behavior broadly (i.e., time spent sitting or reclining with energy expenditure ≤ 1.5 MET), including outcomes such as screen time, electronic media use, and reading time. However, after applying the inclusion criteria, we found that most eligible studies assessed screen time specifically. Therefore, the Population, Exposure, Comparison, and Outcome (PECO) framework was refined to adopt screen time—rather than sedentary behavior in its broader definition—as the primary outcome (Morgan et al., 2018).

The PRISMA statement recommendations for systematic review reporting were followed (Ardern et al., 2021).

Eligibility criteria

The eligibility criteria were based on the PECO strategy (Morgan et al., 2018), considering children and adolescents (P) with ADHD diagnosis, as defined by the original studies, including diagnoses based on clinical assessment or parent report (E), compared to their peers with typical development (C), in relation to screen time (O). Observational studies that addressed all components of the PECO framework were considered eligible. Editorials, letters to the editor, literature reviews, case reports, and case series were excluded, as these designs do not provide empirical comparative data required for this type of review. Observational studies that did not include a comparison group (children and adolescents with typical development) or that assessed exposure and outcome in an adult population were also excluded.

Literature search strategy

The articles were identified through a systematic search of the following electronic databases: MedLine via PubMed, Cochrane Library, EMBASE, EBSCO (CINAHL Database/SPORTDiscus), ProQuest, PsycINFO, Scopus, Web of Science, Lilacs, and the gray literature from Google Scholar and Open Grey. The search strategies defined for the databases described above are listed in Table 1. Two authors (PGJ and IESS) performed the search independently. Subject-matter experts were also consulted to assist in the systematic search. Searches were performed including studies published up to December 2022, with updates given via alerts in the respective databases.

The terms that guided the search were: "sedentary behavior," "lifestyle," "screen time," "lack of physical activity," "lifestyle," "lifestyles," "video games," "computer games," "television viewing," "digital media," "sitting position," "bed rest," "rest," "bedrest," "bedrests," "attention deficit disorder with hyperactivity," "attention deficit hyperactivity disorder," and "attention deficit and disruptive behavioral disorders," along with their synonyms and

Table 1. Search strategy.

Database	Search strategy
PubMed	*("Sedentary Behavior"[MeSH Terms] OR "life style"[MeSH Terms] OR "screen time"[MeSH Terms] OR "lack of physical activity"[Title/Abstract] OR "Lifestyle"[Title/Abstract] OR "Lifestyles"[Title/Abstract] OR ("sitting position"[MeSH Terms] AND ("Time"[MeSH Terms] OR "Time"[Text Word])) OR "bed rest"[MeSH Terms] OR ("Rest"[MeSH Terms] AND "Bedrest"[Title/Abstract]) OR "Bedrests"[Title/Abstract] OR "Sedentary Behavior"[MeSH Terms] OR "bed rest"[Text Word] OR "screen time"[MeSH Terms] OR "screen-based entertainment"[Text Word] OR "screen time"[Text Word] OR ("Television viewing"[Text Word] OR "tv viewing"[Text Word]) OR ("physical*"[Text Word] AND "inactivit*"[Text Word]) OR "Videogames"[mesh terms] OR videogame*[Text Word] OR "Computer Games"[Text Word] OR "digital media"[Text Word]) AND ("attention deficit disorder with hyperactivity"[MeSH Terms] OR "attention deficit hyperactivity disorder"[Title/Abstract] OR ADHD[Title/Abstract] OR "attention deficit and disruptive behavior disorders"[MeSH Terms] OR "attention deficit disorder"[Title/Abstract] OR ("attention deficit"[Title/Abstract] AND "disruptive behavioral disorders"[Title/Abstract]))*
Cochrane Library	*#1MeSH descriptor: [Sedentary Behavior] explode all trees* *#2MeSH descriptor: [Life Style] explode all trees* *#3MeSH descriptor: [Screen Time] explode all trees* *#4MeSH descriptor: [Sitting Position] explode all trees* *#5MeSH descriptor: [Time] explode all trees* *#6MeSH descriptor: [Bed Rest] explode all trees* *#7MeSH descriptor: [Rest] explode all trees* *#8(lack of physical activity OR Lifestyle OR Lifestyles OR Time AND Bedrest OR Bedrests OR bed rest OR screen-based entertainment OR screen time OR Television viewing OR tv viewing OR physical AND inactivit):ti,ab,kw* *#9(videogames OR computer games OR digital media):ti,ab,kw* *#10MeSH descriptor: [Videogames] explode all trees* *#11#1 or #2 or #3 or #4 or #5 or #6 or #7 or #8 or #9 or #10* *#12MeSH descriptor: [Attention Deficit Disorder with Hyperactivity] explode all trees* *#13MeSH descriptor: [Attention Deficit and Disruptive Behavior Disorders] explode all trees* *#14(attention deficit hyperactivity disorder OR attention deficit disorder OR attention deficit AND disruptive behavioral disorders):ti,ab,kw* *#15(ADHD):ti,ab,kw* *#16#12 or #13 or #14 or #15* *#17#11 and #16*
EBSCO (CINAHL Database/ SPORTDiscus)	*(SU "Sedentary Behavior" OR SU "life style" OR SU "screen time" OR SU "sitting position" AND SU "Time" OR SU "bed rest" OR SU "Rest" OR SU "Sedentary Behavior" OR SU "screen time" OR SU "Videogames" OR TI ("lack of physical activity" OR Lifestyle OR Lifestyles OR Bedrest OR Bedrests OR "bed rest" OR "screen-based entertainment" OR "screen time" OR "Television viewing" OR "tv viewing" OR "physical inactivit*" OR "videogame" OR "Computer Games" OR "digital media") OR AB ("lack of physical activity" OR Lifestyle OR Lifestyles OR Bedrest OR Bedrests OR "bed rest" OR "screen-based entertainment" OR "screen time" OR "Television viewing" OR "tv viewing" OR "physical inactivit* OR "videogame" OR "Computer Games" OR "digital media")) AND (SU "attention deficit disorder with hyperactivity" OR SU "attention deficit and disruptive behavior disorders") OR TI ("attention deficit hyperactivity disorder" OR ADHD OR "attention deficit" AND "disruptive behavioral disorders") OR AB ("attention deficit hyperactivity disorder" OR ADHD OR "attention deficit" AND "disruptive behavioral disorders"))*
EMBASE	*("videogame"/mj OR "lifestyle"/mj OR "sedentary lifestyle"/mj OR "screen time"/mj OR "lack of physical activity":ti,ab,kw OR "lifestyle":ti,ab,kw OR "lifestyles":ti,ab,kw OR "time"/mj OR "bed rest"/mj OR "rest"/mj OR "bed rest":ti,ab,kw OR "bed rests":ti,ab,kw OR "screen-based entertainment":ti,ab,kw OR "screen time":ti,ab,kw OR "television viewing":ti,ab,kw OR "videogames":ti,ab,kw OR "computer games":ti,ab,kw OR "digital media":ti,ab,kw) AND ("attention deficit disorder"/mj OR "attention deficit disorder":ti,ab,kw OR "attention deficit and disruptive behavioral disorders":ti,ab,kw OR ADHD:ti,ab,kw)*
LILACS	*((mh:(Sedentary Behavior)) OR (mh:(life style)) OR (mh:(screen time)) OR (mh:(life style)) OR (mh:(sitting position)) OR (mh:(Time)) OR (mh:(bed rest)) OR (mh:(Rest)) OR (mh:(screen time)) OR (lack of physical activity OR Lifestyle OR Lifestyles OR Time AND Bedrest OR Bedrests OR bed rest OR screen-based entertainment OR screen time OR Television viewing OR tv viewing OR physical AND inactivit OR videogames OR computer games OR digital media) OR (mh:(videogames))) AND ((mh:(attention deficit disorder with hyperactivity)) OR (mh:(attention deficit and disruptive behavior disorders)) OR (attention deficit hyperactivity disorder OR attention deficit disorder OR attention deficit AND disruptive behavioral disorders OR ADHD)))*

(*Continued*)

Table 1. (Continued).

Database	Search strategy
Scopus	*((INDEXTERMS("Sedentary Behavior") OR INDEXTERMS("life style") OR INDEXTERMS("screen time") OR TITLE-ABS-KEY("lack of physical activity") OR TITLE-ABS-KEY("LIFESTYLE") OR TITLE-ABS-KEY ("Lifestyles") OR INDEXTERMS("Sitting Position") OR INDEXTERMS("BED REST") OR INDEXTERMS ("REST") OR TITLE-ABS-KEY("BEDREST") OR TITLE-ABS-KEY("BEDRESTS") OR INDEXTERMS ("videogames") OR TITLE-ABS-KEY("videogames") OR TITLE-ABS-KEY("computer games") OR TITLE-ABS-KEY("digital media")) AND (INDEXTERMS("attention deficit disorder with hyperactivity") OR INDEXTERMS("attention deficit hyperactivity disorder") OR TITLE-ABS-KEY("attention deficit hyperactivity disorder") OR TITLE-ABS-KEY("attention deficit AND disruptive behavioral disorders") OR TITLE-ABS-KEY(ADHD))*
Web of Science	*((TS = (Sedentary Behavior) OR TS = (life style) OR TS = (screen time) OR TS = (lack of physical activity) OR TS = (Lifestyle*) OR (TS = (sitting position) and TS = (time)) OR (TS = (rest) and TS = (bedrest)) OR (TS = (Rest) and TS = (Sedentary Behavior)) OR TS = ("videogames") OR TS = ("computer games") OR TS = ("digital media")) AND (TS = (attention deficit disorder with hyperactivity) OR (TS = (attention deficit) and TS = (disruptive behavior disorders)) OR (TS = (attention deficit disorder) AND TS = (disruptive behavioral disorders)) OR TS = (ADHD)))*
PsycINFO	(MeSH: *sedentary lifestyle OR* MeSH: *Life Style OR* MeSH: *Screen Time OR* MeSH: *Bed Rest OR* MeSH: *Rest OR* MeSH: *videogame OR* Title: *"Sedentary Behavior" "OR* Title: *'life style' OR* Title: *screen time" OR* Title: *"physical activity" OR* Title: *lifestyle OR* Title: *"sitting position" AND time OR* Title: *bedrest OR* Title: *"videogames" OR* Title: *"computer games" OR* Title: *"digital media") AND (MeSH: "attention deficit disorder with hyperactivity" OR MeSH: "attention deficit hyperactivity disorder" OR Title: "attention deficit hyperactivity disorder" OR Title: "attention deficit AND disruptive behavioral disorders" OR Title: ADHD)*
Google Scholar	*"Sedentary Behavior" and "attention deficit hyperactivity disorder"*
Open Grey	*Sedentary Behavior and attention deficit hyperactivity disorder*
ProQuest	*(Sedentary Behavior) and (screen time) and (attention deficit hyperactivity disorder)*

MeSH terms. No restrictions were placed on the publication date or language. The search terms were adapted for each database. Articles available in more than one database were considered only once. A manual search was performed in the reference lists of the selected articles.

Study selection

Two authors (PGJ and IESS) independently and manually selected studies by evaluating the titles and abstracts of all studies identified in the searches. Full articles were retrieved and examined when the title and abstract did not provide sufficient information for a definitive decision. An index of identification was created for each eligible study by combining the first author and year of publication. If an article was not located, the authors were contacted via e-mail or institutional contact. The online translator https://www.onlinedoctranslator.com/pt/was used to read studies in languages that the authors did not understand (e.g., Turkish). Any disagreements regarding the eligibility of included studies were resolved by consensus or with the help of a third-party author (RSC).

Quality assessment

The Joanna Briggs Institute (JBI) critical appraisal tools for cross-sectional and case-control studies (https://jbi.global/critical-appraisal-tools) were used to critically evaluate the methodology of the included studies. The JBI critical appraisal checklist contains eight questions

for cross-sectional and 10 questions for case-control studies, with four possible responses: "Yes," "No," "Uncertain," and "Not applicable" (Munn et al., 2020).

The evaluations were independently performed by two authors (PGJ and IESS) and used to make the final decision. A third author (RSC) was consulted in cases of disagreement between the two authors' evaluations.

Data collection process

Two authors (PGJ and IESS) extracted the required information from the included articles independently. The following data were extracted: study characteristics (authors, year of publication, country, study design, number of participants included in the study, and age of participants in years), population characteristics (sex, medication use, and number of participants diagnosed or not diagnosed with ADHD), exposure characteristics (type of diagnosis and comorbidities), screen time assessment instrument, type of screen, and measurement (unit) of screen time. Disagreements between authors were resolved by discussion with a third author (RSC) or by consensus. The authors of the included studies were contacted for further details when relevant data were not reported, or for additional information.

Meta-analysis

Data from the studies were analyzed using RevMan software (Review Manager v. 5.3, The Cochrane Collaboration; Copenhagen, Denmark) to assess the association between screen time and ADHD. Two types of analyses were performed:

- Screen time ≥ 2 hours: The odds ratio (OR), log OR, and standard error (SE), or the number of children and adolescents who used screens for ≥2 hours/day (events) and the total number of children and adolescents assessed with (exposure) and without (comparator) ADHD were included in the software to calculate the OR and 95% confidence interval (95%CI). The adjusted OR was used whenever provided by the study; otherwise, the unadjusted OR was calculated and included in the analysis. When necessary, the effect estimates (ORs) were calculated and converted using the RevMan® software tools. Values of $p < .05$ were considered significant.
- Mean screen time: The mean and standard deviation of time (in hours) using screens per day and the number of children and adolescents assessed in each group (with and without ADHD) were extracted from the studies, and the mean difference (MD) and 95%CI were calculated.

When possible, individual or subgroup analyses considering screen type, the inclusion of medicated or unmedicated children, and weekly time were performed. A random-effects model was used, and heterogeneity was assessed using the I^2 statistic.

Assessment of the certainty of evidence

The certainty of the evidence (certainty in the effect estimates) was determined for each combined outcome in the meta-analysis using the Grading of Recommendations

Assessment, Development, and Evaluation (GRADE) approach. With the GRADE approach, observational studies begin with low evidence; however, this can decrease to very low if serious or very serious issues related to the risk of bias, imprecision, inconsistency, indirect validity, and publication bias are present. The magnitude of the effect and dose-response are not applicable and were judged so as not to alter the certainty of the evidence. The influence of possible confounding factors was judged and may increase the certainty of the evidence (Ryan & Hill, 2016). The presence of age groups including children (zero to 10 years) and adolescents (11 to 19 years) was considered to assess indirect validity. The presence of a "large effect" was considered for OR > 4 or <1 due to the risk of 0.5 being equivalent to a probability of 1; thus, the corresponding OR was measured by the formula OR = RR/(1 − RR) (Schünemann et al., 2019). Thus, the quality of the evidence can range from very low to high.

Results

Study selection

The study selection process is presented in Figure 1 according to the PRISMA guidelines (Ardern et al., 2021). A total of 4,969 references were identified in the databases, with 2,632 remaining after removal due to duplication, ineligibility, and other reasons (e.g., abstracts, books, conferences, and letters to the editor).

After reading the titles and abstracts, the number was reduced to 69 studies. The full texts of these 69 studies were assessed for eligibility and 47 were excluded. Thus, 22 studies were included for data extraction and qualitative synthesis (Basay et al., 2020; Bioulac et al., 2008; Bourchtein et al., 2019; Çöl et al., 2019; Engelhardt et al., 2013; Holton & Nigg, 2020; Hong

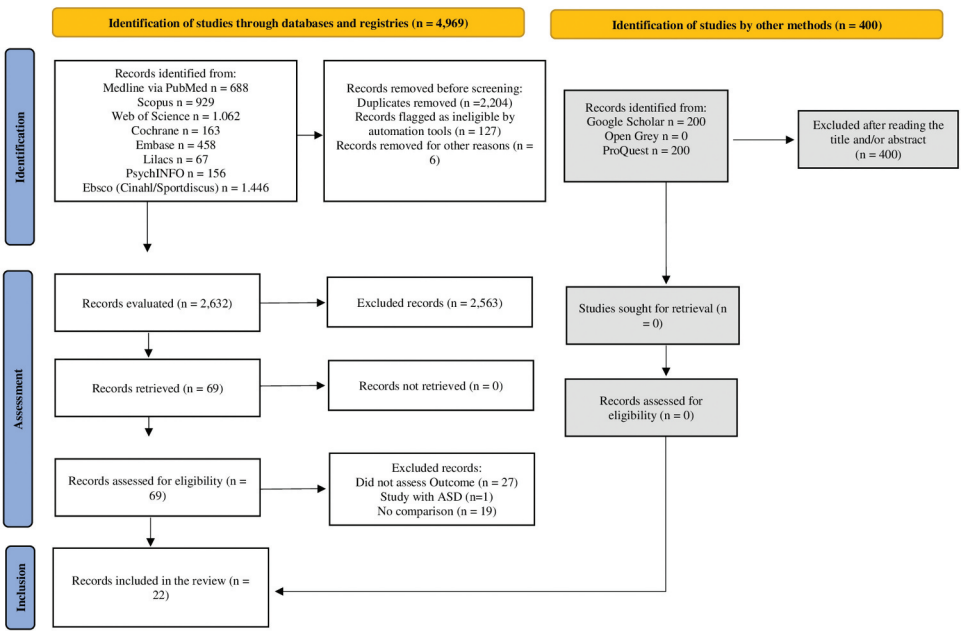


Figure 1. PRISMA flowchart of the database search results. ASD: autism spectrum disorder.

et al., 2021; Khalil et al., 2022; Kietglaiwansiri & Chonchaiya, 2018; Kim et al., 2011; Lingineni et al., 2012; Loewen et al., 2020; Masi et al., 2021; Mazurek & Engelhardt, 2013; Mercurio et al., 2021; Merrill et al., 2021; Párraga et al., 2019; San Mauro Martín et al., 2018; Tandon et al., 2019; Thoma et al., 2020; Tong et al., 2016; Verma et al., 2022) and 13 for quantitative analyses in this review (Bourchtein et al., 2019; Engelhardt et al., 2013; Holton & Nigg, 2020; Kietglaiwansiri & Chonchaiya, 2018; Kim et al., 2011; Loewen et al., 2020; Mazurek & Engelhardt, 2013; Merrill et al., 2021; Párraga et al., 2019; San Mauro Martín et al., 2018; Tandon et al., 2019; Thoma et al., 2020; Tong et al., 2016).

Characteristics of the included articles

The characteristics of the 22 included studies are listed in Table 2. Twenty-one studies presented a cross-sectional design and one a case-control design (San Mauro Martín et al., 2018). Four studies were conducted in Europe (Bioulac et al., 2008; Párraga et al., 2019; San Mauro Martín et al., 2018; Thoma et al., 2020), 11 in North America (Bourchtein et al., 2019; Engelhardt et al., 2013; Holton & Nigg, 2020; Kim et al., 2011; Lingineni et al., 2012; Loewen et al., 2020; Masi et al., 2021; Mazurek & Engelhardt, 2013; Mercurio et al., 2021; Merrill et al., 2021; Tandon et al., 2019), one in Oceania, North America, and Europe (Hong et al., 2021), and one in the Middle East (Khalil et al., 2022). Two studies recruited male children and adolescents only (Engelhardt et al., 2013; Mazurek & Engelhardt, 2013), while the other 20 studies evaluated both genders (Basay et al., 2020; Bioulac et al., 2008; Bourchtein et al., 2019; Çöl et al., 2019; Holton & Nigg, 2020; Hong et al., 2021; Khalil et al., 2022; Kietglaiwansiri & Chonchaiya, 2018; Kim et al., 2011; Lingineni et al., 2012; Loewen et al., 2020; Masi et al., 2021; Mercurio et al., 2021; Merrill et al., 2021; Párraga et al., 2019; San Mauro Martín et al., 2018; Tandon et al., 2019; Thoma et al., 2020; Tong et al., 2016; Verma et al., 2022). The number of participants ranged in these studies from 50 (Bioulac et al., 2008) to 68,634 (Lingineni et al., 2012).

The participants' ages ranged from less than 5 years (Khalil et al., 2022) to 19 years (Merrill et al., 2021). Sixteen studies diagnosed ADHD via clinical examination (Basay et al., 2020; Bioulac et al., 2008; Bourchtein et al., 2019; Çöl et al., 2019; Engelhardt et al., 2013; Holton & Nigg, 2020; Hong et al., 2021; Khalil et al., 2022; Kietglaiwansiri & Chonchaiya, 2018; Loewen et al., 2020; Masi et al., 2021; Mazurek & Engelhardt, 2013; Párraga et al., 2019; San Mauro Martín et al., 2018; Thoma et al., 2020; Verma et al., 2022) or DSM-5 (Basay et al., 2020; Holton & Nigg, 2020; Hong et al., 2021; Thoma et al., 2020). Six studies made diagnoses based on parent, guardian, or caregiver reports (Kim et al., 2011; Lingineni et al., 2012; Mercurio et al., 2021; Merrill et al., 2021; Tandon et al., 2019; Tong et al., 2016).

In the included studies, screen time was predominantly assessed using parent-report questionnaires. A small number of studies employed additional or alternative informants, including guardian reports (Lingineni et al., 2012; Mercurio et al., 2021) or teacher reports (Kietglaiwansiri & Chonchaiya, 2018), and one study used direct assessment from patients using the Robinson Scale (San Mauro Martín et al., 2018). The main screen types evaluated were the following: video games (Bioulac et al., 2008; Bourchtein et al., 2019; Engelhardt et al., 2013; Holton & Nigg, 2020; Kietglaiwansiri & Chonchaiya, 2018; Kim et al., 2011; Lingineni et al., 2012; Masi et al., 2021; Mazurek & Engelhardt, 2013; Párraga et al., 2019; Thoma et al., 2020); computers (Basay et al., 2020; Bourchtein et al., 2019; Çöl et al., 2019; Engelhardt et al., 2013; Lingineni et al., 2012; Loewen et al., 2020; Tandon et al., 2019;

Table 2. Description of included studies.

Author, year, and country	Study design	Sample	Age group	Sex	No ADHD	ADHD
Bioulac et al. (2008) - France	Cross-sectional	50	6 to 16	Both	21	29
Kim et al. (2011) - USA	Cross-sectional	66,643	6 to 17	Both	60,573	6,070
Lingineni et al. (2012) - USA	Cross-sectional	68,634	5 to 17	Both	61,378	7,137
Engelhardt et al. (2013) - USA	Cross-sectional	128	8 to 17	Male	41	38
Mazurek and Engelhardt (2013) - USA	Cross-sectional	141	8 to 18	Male	41	44
Tong et al. (2016) - China	Cross-sectional	785	9 to 13	Both	82	703
San Mauro Martín et al. (2018) - Spain	Case-control	89	8 to 12	Both	48	41
Kietglaiwansiri and Chonchaiya (2018) - Thailand	Cross-sectional	182	8 to 11	Both	102	80
Çöl et al. (2019) - Turkey	Cross-sectional	177	6 to 12	Both	100	77
Párraga et al. (2019) - Spain	Cross-sectional	160	6 to 16	Both	80	80
Tandon et al. (2019) - USA	Cross-sectional	57,513	6 to 18	Both	47,532	4,267
Bourchtein et al. (2019) - USA	Cross-sectional	302	12 to 14	Both	140	162
Basay et al. (2020) - Turkey	Cross-sectional	227	6 to 17	Both	182	55
Holton and Nigg (2020) - USA	Cross-sectional	288	7 to 11	Both	104	184
Loewen et al. (2020) - Canada	Cross-sectional	3,436	10 to 11	Both	3065	371
Thoma et al. (2020) - Germany	Cross-sectional	122	8 to 18	Both	61	61
Hong et al. (2021) - New Zealand, the United States of America, Canada, the United Kingdom, and Australia	Cross-sectional	309	9	Both	255	54
Masi et al. (2021) - Canada	Cross-sectional	280	4 to 12	Both	182	98
Mercurio et al. (2021) - USA	Cross-sectional	34,675	6 to 17	Both	30,176	4,499
Merrill et al. (2021) - USA	Cross-sectional	1,003	6 to 19	Both	826	177
Khalil et al. (2022) - Egypt	Cross-sectional	69	> 5	Both	30	39
Verma et al. (2022) - India	Cross-sectional	70	6 to 16	Both	40	41

Author, year, and country	Diagnosis	Medication	Comorbidity
Bioulac et al. (2008) - France	Clinical	No	Behavioral disorder (oppositional, defiant, and conduct) and emotional disorder (mood and anxiety)
Kim et al. (2011) - USA	Report	Yes	Depression/anxiety
Lingineni et al. (2012) - USA	Report	Yes	Depression/anxiety
Engelhardt et al. (2013) - USA	Clinical	Yes	Autistic Spectrum Disorder
Mazurek and Engelhardt (2013) - USA	Clinical	Yes	Autistic Spectrum Disorder
Tong et al. (2016) – China	Report	No	-
San Mauro Martín et al. (2018) –Spain	Clinical	Yes	-
Kietglaiwansiri and Chonchaiya (2018) - Thailand	Clinical	No	-
Çöl et al. (2019) – Turkey	Clinical	No	-
Párraga et al. (2019) – Spain	Clinical	No	-
Tandon et al. (2019) – USA	Report	Yes	Autism Spectrum Disorder and Asthma
Bourchtein et al. (2019) - USA	Clinical	Yes	Sleep disorder

(*Continued*)

Table 2. (Continued).

Author, year, and country	Diagnosis	Medication	Comorbidity
Basay et al. (2020) – Turkey	Clinical	No	Disorders: conduct, depressive, anxiety, bipolar, psychotic, learning, obsessive-compulsive and tics; disorders: speech, elimination, other
Holton and Nigg (2020) - USA	Clinical	Yes	Disorders: mood, anxiety, conduct, and obsessive-challenging
Loewen et al. (2020) - Canada	Clinical	No	-
Thoma et al. (2020) - Germany	Clinical	No	Sleep disorder
Hong et al. (2020) - New Zealand, the United States of America, Canada, the United Kingdom, and Australia	Clinical	Yes	-
Masi et al. (2021) - Canada	Clinical	No	-
Mercurio et al. (2021) – USA	Report	No	-
Merrill et al. (2021) - USA	Report	Yes	-
Khalil et al. (2022) - Egypt	Clinical	No	-
Verma et al. (2022) - India	Clinical	No	-

ADHD: Attention Deficit Hyperactivity Disorder.

Thoma et al., 2020; Verma et al., 2022); phones, cell phones, and tablets (Basay et al., 2020; Khalil et al., 2022; Thoma et al., 2020; Tong et al., 2016; Verma et al., 2022); movies, reading, and videos (Bourchtein et al., 2019; Holton & Nigg, 2020; Kim et al., 2011; Párraga et al., 2019); and electronic devices, internet, and exposure to digital media (Holton & Nigg, 2020; Hong et al., 2021; Mercurio et al., 2021; San Mauro Martín et al., 2018; Verma et al., 2022). One study did not specify the screen type assessed (Merrill et al., 2021).

Twelve studies did not consider medication use in their design or analysis (Basay et al., 2020; Bioulac et al., 2008; Çöl et al., 2019; Khalil et al., 2022; Kietglaiwansiri & Chonchaiya, 2018; Loewen et al., 2020; Masi et al., 2021; Mercurio et al., 2021; Párraga et al., 2019; Thoma et al., 2020; Tong et al., 2016; Verma et al., 2022) and 10 studies explicitly considered medication use (Bourchtein et al., 2019; Engelhardt et al., 2013; Holton & Nigg, 2020; Hong et al., 2021; Kim et al., 2011; Lingineni et al., 2012; Mazurek & Engelhardt, 2013; Merrill et al., 2021; San Mauro Martín et al., 2018; Tandon et al., 2019).

Regarding the associated comorbidities reported by the studies, the children and adolescents had disorders such as depression (Basay et al., 2020; Kim et al., 2011; Lingineni et al., 2012), anxiety (Basay et al., 2020; Bioulac et al., 2008; Holton & Nigg, 2020; Kim et al., 2011; Lingineni et al., 2012), conduct disorders (Basay et al., 2020; Bioulac et al., 2008; Holton & Nigg, 2020), oppositional defiant disorder (Bioulac et al., 2008), obsessive-compulsive disorder (Basay et al., 2020), mood disorders (Bioulac et al., 2008), bipolar disorder (Basay et al., 2020), psychotic disorders, learning difficulties (e.g., dyslexia), tics, speech disorders, elimination disorders (Basay et al., 2020), sleep disorders (Hong et al., 2021; Thoma et al., 2020), autism spectrum disorders (Engelhardt et al., 2013; Mazurek & Engelhardt, 2013; Tandon et al., 2019), and asthma (Tandon et al., 2019) associated with ADHD. Ten studies treated screen time as a continuous variable (Basay et al., 2020; Bourchtein et al., 2019; Engelhardt et al., 2013; Hong et al., 2021; Masi et al., 2021; Mazurek & Engelhardt, 2013; Merrill et al., 2021; Párraga et al., 2019; Thoma et al., 2020; Verma et al., 2022), eleven treated it as a categorical variable (Bioulac et al., 2008; Çöl et al., 2019; Holton & Nigg, 2020; Khalil et al., 2022; Kietglaiwansiri & Chonchaiya, 2018; Kim et al., 2011; Lingineni et al., 2012; Loewen et al., 2020; Mercurio et al., 2021; San Mauro Martín et al., 2018; Tong et al., 2016), and one considered it both ways (Tong et al., 2016). Among the studies that analyzed screen time as a categorical variable, six operated the variable in a dichotomous manner (Çöl et al., 2019; Holton & Nigg, 2020; Kim et al., 2011; Lingineni et al., 2012; Loewen et al., 2020; Mercurio et al., 2021) and five in a polytomous manner (Bioulac et al., 2008; Khalil et al., 2022; Kietglaiwansiri & Chonchaiya, 2018; Kim et al., 2011; Tong et al., 2016).

Quality assessment

Regarding the quality assessment in the individual studies, 17 studies covered between 75% and 100% of the qualifiers (Basay et al., 2020; Bioulac et al., 2008; Bourchtein et al., 2019; Holton & Nigg, 2020; Hong et al., 2021; Khalil et al., 2022; Kietglaiwansiri & Chonchaiya, 2018; Kim et al., 2011; Lingineni et al., 2012; Loewen et al., 2020; Masi et al., 2021; Mercurio et al., 2021; Merrill et al., 2021; Tandon et al., 2019; Thoma et al., 2020; Tong et al., 2016; Verma et al., 2022), three studies considered between 51% and 74% (Engelhardt et al., 2013; Mazurek & Engelhardt, 2013; Párraga et al., 2019), and two considered 50% of the questions (Çöl et al., 2019; San Mauro Martín et al., 2018). More information is provided in Table 3.

Meta-analysis and GRADE

Five studies were not included in the meta-analysis because they did not provide sufficient data (Basay et al., 2020; Hong et al., 2021; Masi et al., 2021; Merrill et al., 2021; Verma et al., 2022), and one because the sample consisted of children under 5 years of age (Khalil et al., 2022).

- Two hours or more of screen time: The studies had sufficient data only for the analyses considering screen time during the weekdays. Seven studies and eight groups (8,513 children and adolescents with ADHD and 11,961 without ADHD) were included.

The results of the meta-analysis of the seven included studies showed a higher likelihood of screen use for ≥2 hours/day on weekdays among children and adolescents with ADHD, with very low certainty of evidence [OR 1.38 (1.07, 1.77); $p = .001$; $I^2 = 66\%$), compared to their peers with typical development. Subgroup analysis indicated that, in studies evaluating screen time overall, the odds were similar between groups [OR 1.14 (0.94, 1.38); $p = .17$; $I^2 = 46\%$]. However, when only computer use was considered, children and adolescents with ADHD were found to be more likely to use this type of screen for ≥2 hours/day on weekdays [OR 2.61 (1.37, 4.99); $p = .004$; $I^2 = 0\%$] (Figure 2 and Table 4).

Regarding the use or not of medication, children and adolescents with ADHD on medication [OR 1.01 (0.84, 1.21); $p = .93$; $I^2 = 29\%$] showed similar odds of engaging in ≥2 hours/day of screen use compared to their typically developing peers, and children and adolescents off medication [OR 1.06 (0.85, 1.33); $p = .61$; $I^2 = 44\%$] also presented similar odds of ≥2 hours/day of screen use. However, in studies that did not specify medication use in their inclusion criteria or outcomes, a higher likelihood of screen use was observed among children and adolescents with ADHD [OR 1.81 (1.06, 3.09); $p = .03$; $I^2 = 65\%$] (Figure 3). According to the GRADE framework, the certainty of the evidence was considered low in the analyses involving children and adolescents on or off medication, and very low in studies that did not specify medication use (Table 4).

- Daily mean screen time: The analyses are presented below according to the type of screen evaluated.

For overall screen use, three studies were included. On weekdays, children and adolescents with ADHD ($n = 222$) had higher mean screen time than those without ADHD ($n = 201$) [MD = 0.49 (0.18, 0.80); $p = .002$; $I^2 = 54\%$]. However, during weekends, mean screen time was similar between children and adolescents with ADHD ($n = 113$) and controls ($n = 316$) [MD = 0.37 (−0.01, 0.75); $p = .06$; $I^2 = 7\%$] (Figure 4). The certainty of evidence for both analyses was judged as low according to the GRADE approach (Table 5).

For computer use, three studies (4,385 children and adolescents with ADHD and 47 653 without ADHD) were included. The mean computer use time was similar between groups, both in the analysis of weekdays [MD = 0.03 (−0.29, 0.35); $p = .86$; $I^2 = 84\%$] and in studies including medicated participants [MD = 0.20 (−0.51, 0.92); $p = .56$; $I^2 = 89\%$]. Overall, the mean time spent using computers did not differ significantly between groups [MD = 0.17

Table 3. JBI critical assessment checklist for the cross-sectional and case control studies.

Authors	1^{CB}	2^{CB}	3^{CB}	4^{CB}	5^{CB}	6^{CB}	7^{CB}	8^{CB}	9^{B}	10^{B}	Score	Overall Quality Results
Bioulac et al. (2008)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Unclear	–	–	87.5%	7/8
Kim et al. (2011)	Yes	Yes	No	Yes	Yes	Yes	Yes	Yes	–	–	87.5%	7/8
Lingineni et al. (2012)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	–	–	100.0%	8/8
Engelhardt et al. (2013)	Yes	Unclear	Yes	No	Yes	Yes	No	Yes	–	–	62.5%	5/8
Mazurek and Engelhardt (2013)	Yes	Unclear	Yes	No	Yes	Yes	No	Yes	–	–	62.5%	5/8
Tong et al. (2016)	Yes	Unclear	Yes	Yes	Yes	Yes	Yes	Yes	–	–	87.5%	7/8
San Mauro Martín et al. (2018)	No	No	Yes	Yes	Yes	Yes	No	Yes	No	No	50.0%	5/10
Kietglaiwansiri and Chonchaiya (2018)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	–	–	80.0%	8/8
Çöl et al. (2019)	Yes	Yes	Yes	No	Yes	No	No	No	–	–	50.0%	4/8
Párraga et al. (2019)	No	No	Yes	Yes	Yes	Unclear	Yes	Yes	–	–	62.5%	5/8
Tandon et al. (2019)	Yes	Yes	No	Yes	Yes	Yes	Yes	Yes	–	–	87.5%	7/8
Bourchtein et al. (2019)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	–	–	100.0%	8/8
Basay et al. (2020)	Yes	Yes	Yes	Yes	Yes	Yes	No	No	–	–	75.0%	6/8
Holton and Nigg (2020)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	–	–	100.0%	8/8
Loewen et al. (2020)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	–	–	100.0%	8/8
Thoma et al. (2020)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	–	–	100.0%	8/8
Hong et al. (2021)	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes	–	–	87.5%	7/8
Masi et al. (2021)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	–	–	100.0%	8/8
Mercurio et al. (2021)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	–	–	87.5%	7/8
Merrill et al. (2021)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	–	–	100.0%	8/8
Khalil et al. (2022)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes			100.0%	8/8
Verma et al. (2022)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes			100.0%	8/8
Number of Yes considering the 22 included studies	20/22	17/22	20/22	19/22	22/22	20/22	17/22	20/22	0/1	0/1	–	–

[C]questions applied to crosssectional JBI, [B]questions applied to JBI for case control studies.

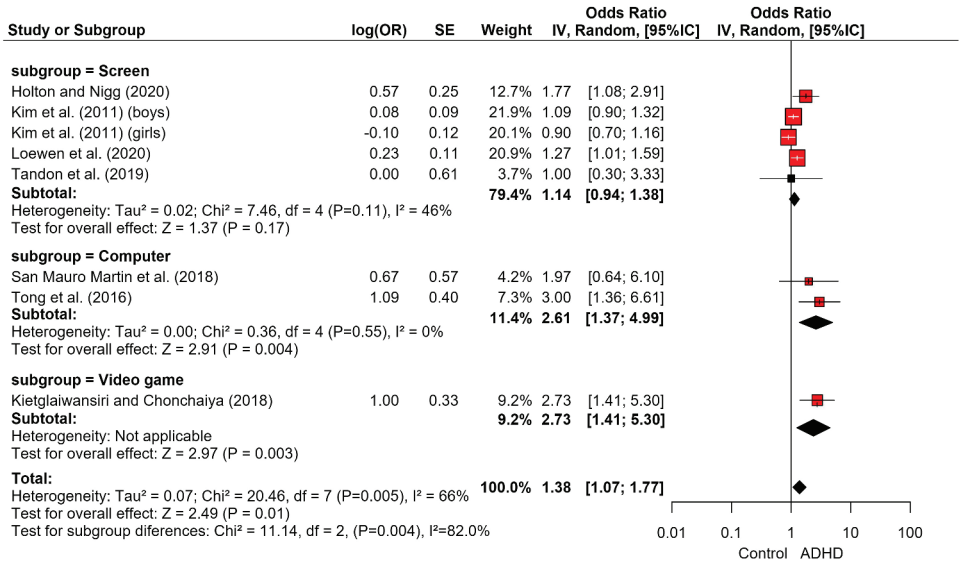


Figure 2. Chance of using screen for ≥2 hours during weekdays according to screen type.

(−0.18, 0.52); $p = .33$; $I^2 = 84\%$] (Figure 5). The certainty of the evidence was rated as very low (Table 5).

For television use, four studies (4,547 children and adolescents with ADHD and 47 793 without ADHD) were included. The mean television use was similar between groups on weekdays [MD = 0.00 (−0.33, 0.33); $p = 1.00$; $I^2 = 81\%$] and in studies with medicated participants [MD = −0.15 (−0.58, 0.28); $p = .49$; $I^2 = 71\%$]. Overall, there was no significant difference in mean television use between the groups [MD = 0.02 (−0.25, 0.29); $p = .88$; $I^2 = 81\%$] (Figure 6). According to GRADE, the certainty of this evidence was very low (Table 5).

For video game use, three studies (286 children and adolescents with medicated and unmedicated ADHD and 261 without ADHD) were included. Children and adolescents with and without ADHD had similar mean video game use [MD = 0.22 (−0.40, 0.84); $p = .48$, $I^2 = 95\%$] (Figure 7). The certainty of evidence was also rated as very low using the GRADE approach (Table 5).

Discussion

The evidence available in the literature raises questions about whether children and adolescents with ADHD are more or less exposed to screen-based sedentary behavior than their typically developing peers (Ebenegger et al., 2012; Párraga et al., 2019). This systematic review and meta-analysis suggests that children and adolescents with ADHD spend more time during weekdays using screens and are more exposed to excessive screen time (i.e., ≥2 hours/day) overall and in relation to computer use. These results do not seem to be influenced by medication use in ADHD patients.

The American Academy of Pediatrics advises that children aged 2 to 18 years should not exceed 2 hours of screen use per day (Reid Chassiakos et al., 2016), while the World Health

Table 4. Certainty of evidence from analyses with dichotomous data.

Certainty assessment								Summary of findings	
Studies	Risk of bias	Inconsistency	Indirectness	Imprecision	Publication bias	Overall certainty of evidence	Relative effect (95% CI)	Anticipated absolute effects	
								Risk with CONTROLS	Risk difference with ADHD
Chance of screen use for two hours or more, during weekdays, according to the type of screen.									
7 observational studies	not serious	serious[a]	not serious	serious[b]	none	⨁◯◯◯ Very low	**OR 1.38** (1.07 to 1.77)	0 per 1000	**1 fewer per 1000** (from 2 fewer to 1 fewer)
Chance of screen use for two hours or more during weekdays in medicated children and adolescents.									
1 observational studies	not serious	not serious	not serious	not serious	none	⨁⨁◯◯ Low	**OR 1.01** (0.84 to 1.21)	0 per 1000	**1 fewer per 1000** (from 1 fewer to 1 fewer)
Chance of screen use for two hours or more during weekdays in unmedicated children and adolescents									
2 observational studies	not serious	not serious	not serious	not serious	none	⨁⨁◯◯ Low	**OR 1.06** (0.85 to 1.33)	0 per 1000	**1 fewer per 1000** (from 1 fewer to 1 fewer)
Chance of screen use for two hours or more, during weekdays in studies that did not specify the presence of medication.									
4 observational studies	not serious	serious[a]	not serious	serious[b]	none	⨁◯◯◯ Very low	**OR 1.81** (1.06 to 3.09)	0 per 1000	**2 fewer per 1000** (from 3 fewer to 1 fewer)

ADHD: attention deficit hyperactivity disorder; CI: Confidence Interval; OR: Odds Ratio.
[a]Significant heterogeneity; [b]Upper and/or lower interval of the confidence interval is greater than 25% of the estimated effect.

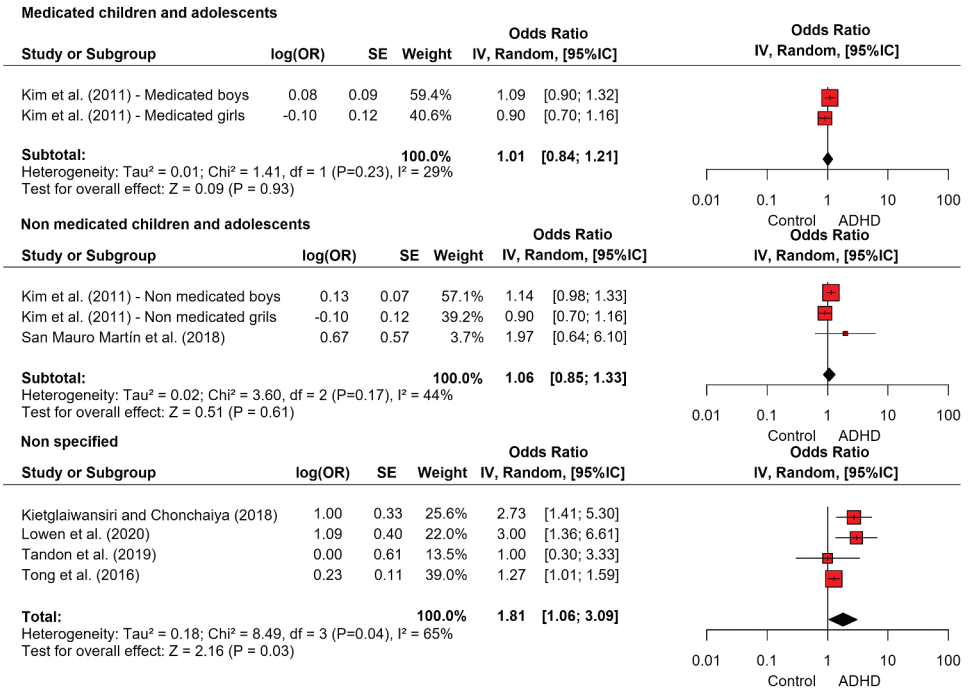


Figure 3. Chance of using screen for ≥2 hours during weekdays according to the presence of medication or not in the case group.

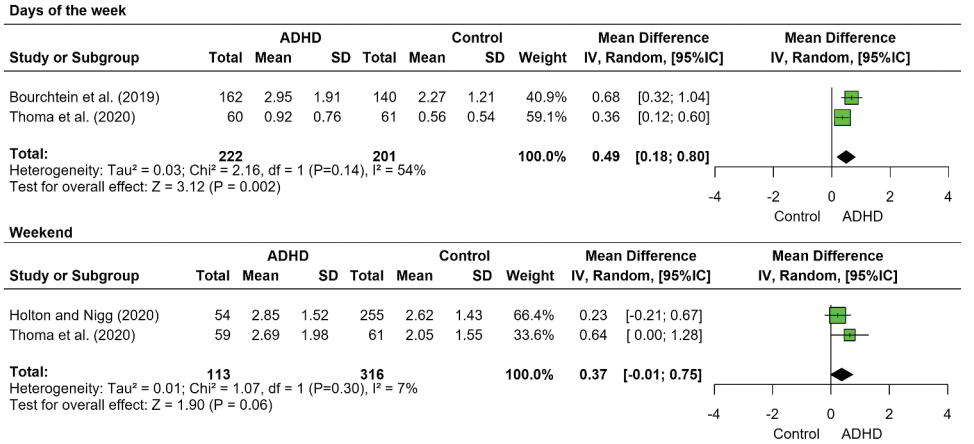


Figure 4. Mean screen time of children with and without ADHD.

Organization states that sedentary behavior, especially time spent on a screen (computer, cell phone, tablet, video games, and television), be reduced as much as possible in children and adolescents aged 5 to 17 years (Bull et al., 2020). Thus, the cutoff time of 2 hours of screen time was used in the analyses with dichotomous data. Children and adolescents with ADHD were more likely to use screens for ≥2 hours/day than the control group.

Table 5. Certainty of the evidence for continuous data.

Certainty assessment							Summary of findings		
							Study event rates (%)		Anticipated absolute effects
Participants (studies)	Risk of bias	Inconsistency	Indirectness	Imprecision	Publication bias	Overall certainty of evidence	Control	ADHD	Risk difference with ADHD
Average hours playing, and screen time overall, by children with and without ADHD during weekdays.									
423	not serious	not serious	not serious	not serious	none	⨁⨁◯◯	201	222	MD **0.49 higher**
(2 observational studies)						Low			(0.18 higher to 0.80 higher)
Average hours playing, screen time in general, by children with and without ADHD over the weekend.									
429	not serious	not serious	not serious	not serious	none	⨁⨁◯◯	316	113	MD **0.37 higher**
(2 observational studies)						Low			(0.01 lower to 0.75 higher)
Average hours of computer use by children with and without ADHD.									
52038	not serious	serious[a]	not serious	not serious	none	⨁◯◯◯	47653	4385	MD **0.17 higher**
(3 observational studies)						Very low			(0.18 lower to 0.52 higher)
Average hours of television use by children with and without ADHD.									
52340	not serious	serious[a]	not serious	not serious	none	⨁◯◯◯	47793	4547	MD **0.02 higher**
(4 observational studies)						Very low			(0.25 lower to 0.29 higher)
Average hours of video game playing by children with and without ADHD.									
547	not serious	very serious[a,b]	not serious	serious[c]	none	⨁◯◯◯	261	286	MD **0.22 higher**
(3 observational studies)						Very low			(0.4 lower to 0.84 higher)

ADHD: attention deficit hyperactivity disorder; MD: mean difference.
[a]Significant heterogeneity. [b]Different directions among the effects of the studies. [c]Upper and/or lower confidence interval is greater than 0.5.

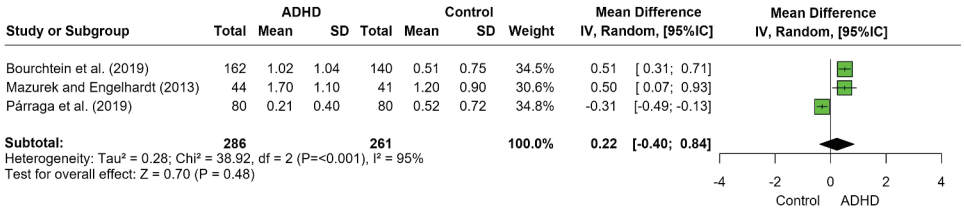


Figure 5. Mean hours of computer use by children with and without ADHD.

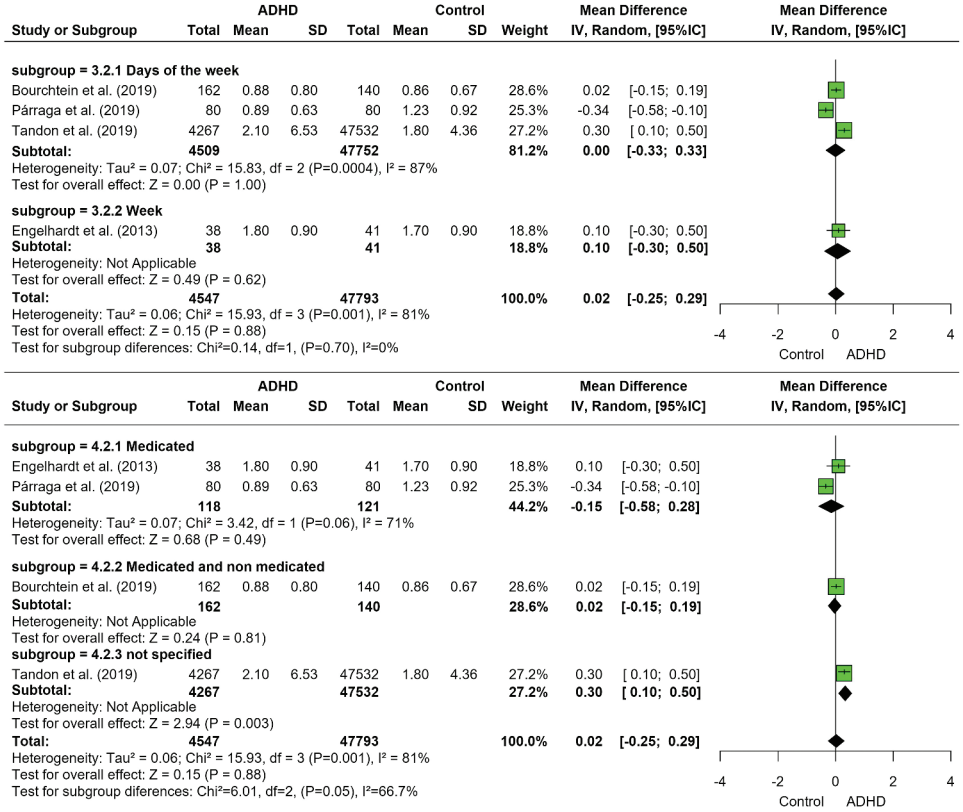


Figure 6. Mean hours of television use by children with and without ADHD.

A positive association was observed between a screen time of ≥2 hours/day on the computer and ADHD (subgroup analysis). However, this difference was not observed for the analysis with continuous data, where children and adolescents with and without ADHD had similar mean computer use time. The conflict in the results may be associated with the difficulty of eliminating individual factors that may interfere with the results, such as age, gender, social context, and relationship with parents, among others. In the dichotomous analyses, adjusted OR data were entered into the analyses whenever they were available in the included studies. However, only two studies (Kim et al., 2011; Tandon et al., 2019), presented adjusted data, indicating the need for further research that considers confounding factors to confirm or ratify this result.

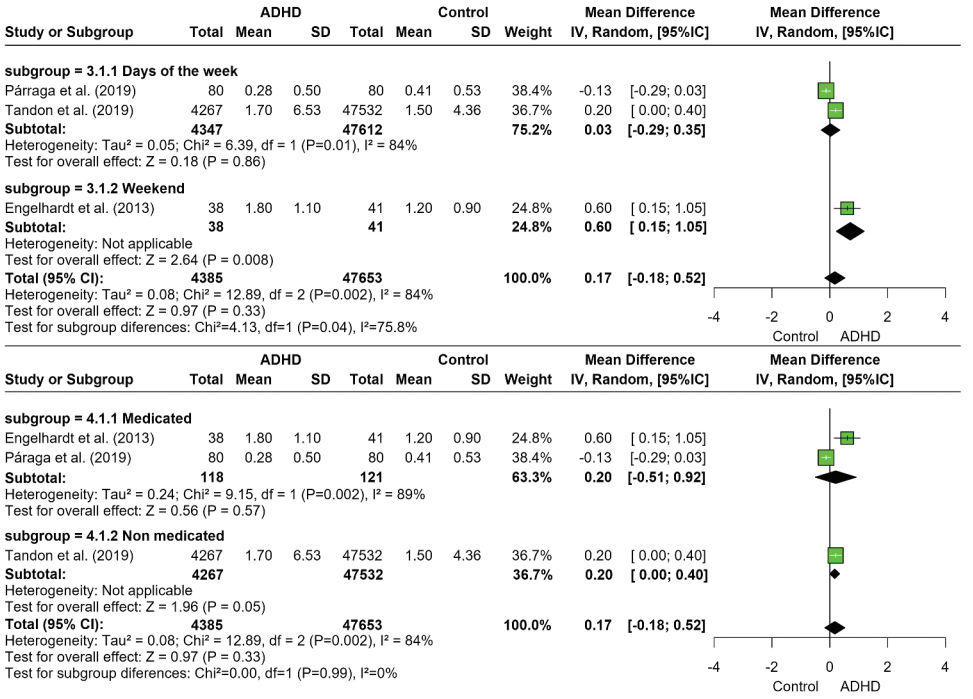


Figure 7. Mean hours playing video games by children with and without ADHD.

When considering screen types individually, the mean time spent using computers, television, and video games was similar among children and adolescents with and without ADHD. It should be noted that, in most analyses, the daily mean usage exceeded 1 hour for each medium. Although these may be considered low values, several types of media are usually used by children during the day (Bourchtein et al., 2019; Párraga et al., 2019), suggesting that the sum of screen time on different media may reach values considered excessive. Most of the studies included in this review were conducted in high-income countries, which may be associated with the higher media use in this population, as there is a significant global digital divide in Internet access, which is substantially higher in high-income nations compared to low-middle income countries (International Telecommunication Union [ITU], 2023).

Only one included study (Khalil et al., 2022) was conducted with a sample of children younger than 5 years old and demonstrated an association between screen use and ADHD. This result should be interpreted with caution, as ADHD is a neurodevelopmental condition and not caused by screen time. Although excessive screen time may negatively impact attention and self-regulation, this is distinct from a clinical diagnosis of ADHD. Guidance should be provided to parents, educators, healthcare professionals, and the community at large about the harmful physiological and psychological health-related effects of excessive electronic media use on children and adolescents. Recommendations, educational campaigns, and public policies to reduce screen use among children and adolescents may be a way forward, as exemplified by the Common Sense organization, promoter of the Digital Detox

campaign, which guides families on the healthy and safe use of technology and encourages children and adolescents to spend more time on offline activities (Syvertsen, 2020).

Medication is an important element of therapeutic strategies for ADHD; in this scenario, the specific medication, daily dosage, and duration of use may act as moderating factors in this association. However, this is a complex topic that involves variability in medication types, dosing strategies, and individual responses. Due to its breadth, it was not possible to explore these factors in depth in the present review. It is reported that adverse effects related to ADHD medication use are less evident in youth when it is taken for longer periods (Becker & Lienesch, 2018). This factor may be related to the chance of using screens for ≥2 hours/day being similar in the analyses that included the studies with both medicated and unmedicated patients; however, children and adolescents with ADHD were more likely to use screens in the studies that did not specify medication use. Future studies evaluating the influence of different medications, as well as their dosages and duration of use, on sleep, media use, and sedentary behavior may shed more light on this issue.

In addition, considering the broad developmental and technological changes in childhood and adolescence, it may be difficult to isolate ADHD as a primary contributing factor to screen time patterns. This challenge highlights the potential value of qualitative research to explore individual experiences, motivations, and contexts that influence screen use behaviors among this population.

It is also essential to acknowledge the potential positive uses of screen-based technologies for children and adolescents with ADHD. Screens are not exclusively a source of sedentary behavior or distraction; they can be powerful tools for learning, cognitive training, and self-regulation. For instance, certain educational apps and structured digital platforms can leverage immediate feedback, novelty, and interactive elements to improve executive functions and attention skills in ways that traditional methods may not (Löchner et al., 2025). Furthermore, video communication and online communities can facilitate crucial social engagement and belonging, particularly for individuals who may struggle with face-to-face social cues due to their condition (Radesky & Christakis, 2016). Thus, the focus should not solely be on minimizing time, but also on promoting high-quality, active, and productive screen engagement (Kollins et al., 2020).

It should be kept in mind that the digital age is a reality and digital game-based therapies may have the potential to treat ADHD, as computer use can improve attention in some children and adolescents by addressing neurocognitive domains with mild side effects (Pandian et al., 2021). However, there is an urgent need to develop measurement approaches regarding children's media use in a non-educational and non-therapeutic manner. Physical activities have several benefits for children with ADHD and may reduce their ADHD symptoms; thus, such activities should be encouraged in this population (Vysniauske et al., 2020). Exergames (i.e., video games that require physical exertion or movement from the player) can also be a tool in this context, merging media and physical activity (Benzing & Schmidt, 2019).

A higher chance of excessive computer use among children and adolescents with ADHD was observed compared to the control group. No study included in the meta-analysis evaluated screen time durations of ≥2 hours/day on weekends. According to Sigmundová and Sigmund (2021), weekends are characterized by more recreational screen time for all family members (parents and children) relative to weekdays.

Studies that evaluate screen use for ≥2 hours/day during weekends are needed to elucidate this issue.

It is important to highlight that the outcomes assessed in this systematic review were classified with low or very low evidence certainty, indicating that the confidence in the estimated effect found in the meta-analysis is limited and that the true effect may be substantially different from the estimated effect, reinforcing the need for new studies on the subject with appropriate methodological designs. Although this review included 22 studies, the evidence from these, in compilation, still does not prove robust.

A limitation is that screen time was assessed in the studies through questionnaires answered by parents, caregivers, and teachers, which may have resulted in response bias. In some studies, ADHD diagnosis was also based on parent report rather than clinical confirmation, which may introduce additional misclassification bias. Objective methods for assessing screen time, such as direct observation, the use of cameras/filming, and electronic applications should be considered in future studies. Another limitation is that several studies did not clearly specify whether screen time included use during school hours or only out-of-school exposure, which may affect comparability across age groups and weekdays. Finally, the search strategy covered studies published up to 2022, and more recent evidence may therefore not be captured in this review.

In conclusion, this systematic review and meta-analysis suggests with low or very low evidence certainty that compared to their typically developing peers, children and adolescents with ADHD (i) spend more time during weekdays on screen use and (ii) are more likely to have an excessive screen and computer time in general. (iii) Medication use in ADHD patients does not seem to influence these findings. These results should be interpreted with caution, and further studies are needed.

Disclosure statement

No potential conflict of interest was reported by the author(s).

References

American Psychiatric Association. (2014). *DSM-5 - manual diagnóstico e estatístico de transtornos mentais* (5a edição ed.). Artmed.

Arango-Tobón, O. E., Guevara Solórzano, A., Orejarena Serrano, S. J., & Olivera La Rosa, A. (2023). Social cognition and prosocial behavior in children with attention deficit hyperactivity disorder: A systematic review. *Healthcare*, *11*(10), 1366. https://doi.org/10.3390/healthcare11101366

Ardern, C. L., Büttner, F., Andrade, R., Weir, A., Ashe, M. C., Holden, S., Impellizzeri, F. M., Delahunt, E., Dijkstra, H. P., Mathieson, S., Rathleff, M. S., Reurink, G., Sherrington, C., Stamatakis, E., Vicenzino, B., Whittaker, J. L., Wright, A. A., Clarke, M., Moher, D., … Winters, M. (2021). Implementing the 27 PRISMA 2020 statement items for systematic reviews in the sport and exercise medicine, musculoskeletal rehabilitation and sports science fields: The PERSiST (implementing PRISMA in exercise, rehabilitation, sport medicine and sports science) guidance. *British Journal of Sports Medicine*, *56*(4), 175–195. https://doi.org/10.1136/bjsports-2021-103987

Ayano, G., Demelash, S., Gizachew, Y., Tsegay, L., & Alati, R. (2023). The global prevalence of attention deficit hyperactivity disorder in children and adolescents: An umbrella review of meta-analyses. *Journal of Affective Disorders*, *339*, 860–866. https://doi.org/10.1016/j.jad.2023.07.071

Basay, K., Basay, O., Akdogan, C., Karaisli, S., Satilmis, M., Gozen, B., & Sekerci, B. (2020). Screen use habits among children and adolescents with psychiatric disorders: A cross-sectional study from Turkey. *Psihologija*, *53*(3), 255–271. https://doi.org/10.2298/PSI190802009K

Becker, S. P., & Lienesch, J. A. (2018). Nighttime media use in adolescents with ADHD: Links to sleep problems and internalizing symptoms. *Sleep Medicine*, *51*, 171–178. https://doi.org/10.1016/j.sleep.2018.06.021

Benzing, V., & Schmidt, M. (2019). The effect of exergaming on executive functions in children with ADHD: A randomized clinical trial. *Scandinavian Journal of Medicine and Science in Sports*, *29*(8), 1243–1253. https://doi.org/10.1111/sms.13446

Bioulac, S., Arfi, L., & Bouvard, M. P. (2008). Attention deficit/hyperactivity disorder and video games: A comparative study of hyperactive and control children. *European Psychiatry*, *23*(2), 134–141. https://doi.org/10.1016/j.eurpsy.2007.11.002

Bourchtein, E., Langberg, J. M., Cusick, C. N., Breaux, R. P., Smith, Z. R., & Becker, S. P. (2019). Featured article: Technology use and sleep in adolescents with and without attention-deficit/hyperactivity disorder. *Journal of Pediatric Psychology*, *44*(5), 517–526. https://doi.org/10.1093/jpepsy/jsy101

Bull, F. C., Al-Ansari, S. S., Biddle, S., Borodulin, K., Buman, M. P., Cardon, G., Carty, C., Chaput, J. P., Chastin, S., Chou, R., Dempsey, P. C., DiPietro, L., Ekelund, U., Firth, J., Friedenreich, C. M., Garcia, L., Gichu, M., Jago, R., Katzmarzyk, P. T., … Willumsen, J. F. (2020). World Health Organization 2020 guidelines on physical activity and sedentary behaviour. *British Journal of Sports Medicine*, *54*(24), 1451–1462. https://doi.org/10.1136/bjsports-2020-102955

Çöl, N., Gökçen, C., Kılıç, B., & Karadağ, M. (2019). Prevalence of obesity/hypertension in children and adolescents with ADHD and evaluation of total body composition. *Alpha Psychiatry*, *20*(1), 93–100. https://doi.org/10.5455/apd.298929

Dona, S. W. A., Badloe, N., Sciberras, E., Gold, L., Coghill, D., & Le, H. N. D. (2023). The impact of childhood attention-deficit/hyperactivity disorder (ADHD) on children's health-related quality of life: A systematic review and meta-analysis. *Journal of Attention Disorders*, *27*(6), 598–611. https://doi.org/10.1177/10870547231155438

Ebenegger, V., Marques-Vidal, P. M., Munsch, S., Quartier, V., Nydegger, A., Barral, J., Hartmann, T., Dubnov-Raz, G., Kriemler, S., & Puder, J. J. (2012). Relationship of hyperactivity/inattention with adiposity and lifestyle characteristics in preschool children. *The Journal of Child Neurology*, *27*(7), 852–858. https://doi.org/10.1177/0883073811428009

Engelhardt, C. R., Mazurek, M. O., & Sohl, K. (2013). Media use and sleep among boys with autism spectrum disorder, ADHD, or typical development. *Pediatrics*, *132*(6), 1081–1089. https://doi.org/10.1542/peds.2013-2066

Fujiwara, H., Tsurumi, K., Shibata, M., Kobayashi, K., Miyagi, T., Ueno, T., Oishi, N., & Murai, T. (2022). Life habits and mental health: Behavioural addiction, health benefits of daily habits, and the reward system. *Frontiers in Psychiatry*, *13*, 813507. https://doi.org/10.3389/fpsyt.2022.813507

Holton, K. F., & Nigg, J. T. (2020). The association of lifestyle factors and ADHD in children. *Journal of Attention Disorders*, *24*(11), 1511–1520. https://doi.org/10.1177/1087054716646452

Hong, G. C. C., Conduit, R., Wong, J., DiBenedetto, M., & Lee, E. (2021). Diet, physical activity, and screen time to sleep better: Multiple mediation analysis of lifestyle factors in school-aged children with and without attention deficit hyperactivity disorder. *Journal of Attention Disorders*, *25*(13), 1847–1858. https://doi.org/10.1177/1087054720940417

International Telecommunication Union. (2023). *Measuring digital development: Facts and figures 2023*. https://digitallibrary.un.org/record/4074376?v=pdf

Katzmarzyk, P. T., Powell, K. E., Jakicic, J. M., Troiano, R. P., Piercy, K., & Tennant, B. (2019). Sedentary behavior and health: Update from the 2018 physical activity guidelines advisory committee. *Medicine and Science in Sports and Exercise*, *51*(6), 1227–1241. https://doi.org/10.1249/MSS.0000000000001935

Khalil, A. H., Zayed, A. M., Amer, A., & Baz, H. (2022). Relation between exposure to different multimedia programs and presentation of attention deficit hyperactivity disorder. *Egyptian Journal of Otolaryngology*, *38*(5). https://doi.org/10.1186/s43163-021-00197-4

Kietglaiwansiri, T., & Chonchaiya, W. (2018). Pattern of video game use in children with attention-deficit-hyperactivity disorder and typical development. *Pediatrics International*, *60*(6), 523–528. https://doi.org/10.1111/ped.13564

Kim, J., Mutyala, B., Agiovlasitis, S., & Fernhall, B. (2011). Health behaviors and obesity among US children with attention deficit hyperactivity disorder by gender and medication use. *Preventive Medicine*, *52*(3–4), 218–222. https://doi.org/10.1016/j.ypmed.2011.01.003

Kollins, S. H., DeLoss, D. J., Cañadas, E., Lutz, J., Findling, R. L., Keefe, R. S., Epstein, J. N., Cutler, A. J., & Faraone, S. V. (2020). A novel digital intervention for actively reducing severity of paediatric ADHD (STARS-ADHD): A randomised controlled trial. *Lancet Digital Health*, *2*(4), e168–e178. https://doi.org/10.1016/S2589-7500(20)30017-0

Lingineni, R. K., Biswas, S., Ahmad, N., Jackson, B. E., Bae, S., & Singh, K. P. (2012). Factors associated with attention deficit/hyperactivity disorder among US children: Results from a national survey. *BMC Pediatrics*, *12*(1), 50. https://doi.org/10.1186/1471-2431-12-50

Löchner, J., Carlbring, P., Schuller, B., Torous, J., & Sander, L. B. (2025). Digital interventions in mental health: An overview and future perspectives. *Internet Interventions*, *40*, 100824. https://doi.org/10.1016/j.invent.2025.100824

Loewen, O. K., Maximova, K., Ekwaru, J. P., Asbridge, M., Ohinmaa, A., & Veugelers, P. J. (2020). Adherence to life-style recommendations and attention-deficit/hyperactivity disorder: A population-based study of children aged 10 to 11 years. *Psychosomatic Medicine*, *82*(3), 305–315. https://doi.org/10.1097/PSY.0000000000000787

Mahumud, R. A., Sahle, B. W., Owusu-Addo, E., Chen, W., Morton, R. L., & Renzaho, A. M. N. (2021). Association of dietary intake, physical activity, and sedentary behaviours with overweight and obesity among 282,213 adolescents in 89 low and middle income to high-income countries. *International Journal of Obesity*, *45*(11), 2404–2418. https://doi.org/10.1038/s41366-021-00908-0

Masi, L., Abadie, P., Herba, C., Emond, M., Gingras, M. P., & Amor, L. B. (2021). Video games in ADHD and non-ADHD children: Modalities of use and association with ADHD symptoms. *Frontiers in Pediatrics*, *9*, 632272. https://doi.org/10.3389/fped.2021.632272

Mazurek, M. O., & Engelhardt, C. R. (2013). Video game use in boys with autism spectrum disorder, ADHD, or typical development. *Pediatrics*, *132*(2), 260–266. https://doi.org/10.1542/peds.2012-3956

Mercurio, L. Y., Amanullah, S., Gill, N., & Gjelsvik, A. (2021). Children with ADHD engage in less physical activity. *Journal of Attention Disorders*, *25*(8), 1187–1195. https://doi.org/10.1177/1087054719887789

Merrill, B. M., Morrow, A. S., Sarver, D., Sandridge, S., & Lim, C. S. (2021). Prevalence and correlates of attention-deficit hyperactivity disorder in a diverse, treatment-seeking pediatric overweight/obesity sample. *Journal of Developmental & Behavioral Pediatrics*, *42*(6), 433–441. https://doi.org/10.1097/DBP.0000000000000910

Morgan, R. L., Whaley, P., Thayer, K. A., & Schünemann, H. J. (2018). Identifying the PECO: A framework for formulating good questions to explore the association of environmental and other exposures with health outcomes. *Environment International*, *121*(Pt 1), 1027–1031. https://doi.org/10.1016/j.envint.2018.07.015

Munn, Z., Barker, T. H., Moola, S., Tufanaru, C., Stern, C., McArthur, A., Stephenson, M., & Aromataris, E. (2020). Methodological quality of case series studies: An introduction to the JBI critical appraisal tool. *JBI Evidence Synthesis*, *18*(10), 2127–2133. https://doi.org/10.11124/JBISRIR-D-19-00099

Musálek, M., Scharoun Benson, S. M., Lejcarova, A., & Bryden, P. J. (2020). Cross-lateralisation in children with attention-deficit/hyperactivity disorder and motor skill performance. *The International Journal of Psychology*, *55*(6), 973–982. https://doi.org/10.1002/ijop.12658

Pandian, G. S. B., Jain, A., Raza, Q., & Sahu, K. K. (2021). Digital health interventions (DHI) for the treatment of attention deficit hyperactivity disorder (ADHD) in children - a comparative review of literature among various treatment and DHI. *Psychiatry Research*, *297*, 113742. https://doi.org/10.1016/j.psychres.2021.113742

Párraga, J. L., Perez, B. C., Lopez-Martin, S., Albert, J., Fernandez-Mayoralas, D. M., Fernandez-Perrone, A. L., de Domingo, A. J., Tirado, P., Lopez-Arribas, S., Suarez-Guinea, R., & Fernandez-

Jaen, A. (2019). Attention-deficit/hyperactivity disorder and lifestyle habits in children and adolescents. *Actas Espanolas de Psiquiatria*, *47*(4), 158–164.

Radesky, J. S., & Christakis, D. A. (2016). Increased screen time: Implications for early childhood development and behavior. *Pediatric Clinics*, *63*(5), 827–839. https://doi.org/10.1016/j.pcl.2016.06.006

Reid Chassiakos, Y. L., Radesky, J., Christakis, D., Moreno, M. A., Cross, C., & Council on Communications and Media.. (2016). Children and adolescents and digital media. *Pediatrics*, *138*(5), e20162593. https://doi.org/10.1542/peds.2016-2593

Ryan, R., & Hill, S. (2016). *How to grade the quality of the evidence*. Cochrane Consumers and Communication Group. http://cccrg.cochrane.org/author-resources

San Mauro Martín, I., Blumenfeld Olivares, J. A., Garicano Vilar, E., Echeverry López, M., García Bernat, M., Quevedo Santos, Y., Blanco López, M., Elortegui Pascual, P., Borregon Rivilla, E., & Rincón Barrado, M. (2018). Nutritional and environmental factors in attention-deficit hyperactivity disorder (ADHD): A cross-sectional study. *Nutritional Neuroscience*, *21*(9), 641–647. https://doi.org/10.1080/1028415X.2017.1331952

Saunders, T. J., & Vallance, J. K. (2017). Screen time and health indicators among children and youth: Current evidence, limitations and future directions. *Applied Health Economics and Health Policy*, *15*(3), 323–331. https://doi.org/10.1007/s40258-016-0289-3

Schünemann, H. J., Cuello, C., Akl, E. A., Mustafa, R. A., Meerpohl, J. J., Thayer, K., Morgan, R. L., Gartlehner, G., Kunz, R., Katikireddi, S. V., Sterne, J., Higgins, J. P., Guyatt, G., & GRADE Working Group. (2019). Grade guidelines: 18. How ROBINS-I and other tools to assess risk of bias in nonrandomized studies should be used to rate the certainty of a body of evidence. *Journal of Clinical Epidemiology*, *111*, 105–114. https://doi.org/10.1016/j.jclinepi.2018.01.012

Sigmundová, D., & Sigmund, E. (2021). Weekday-weekend sedentary behavior and recreational screen time patterns in families with preschoolers, schoolchildren, and adolescents: Cross-sectional three cohort study. *International Journal of Environmental Research and Public Health*, *18*(9), 4532. https://doi.org/10.3390/ijerph18094532

Syvertsen, T. (2020). *Digital detox: The politics of disconnecting*. Emerald Publishing Limited. https://doi.org/10.1108/9781787693395

Tandon, P. S., Sasser, T., Gonzalez, E. S., Whitlock, K. B., Christakis, D. A., & Stein, M. A. (2019). Physical activity, screen time, and sleep in children with ADHD. *Journal of Physical Activity and Health*, *16*(6), 416–422. https://doi.org/10.1123/jpah.2018-0215

Thoma, V. K., Schulz-Zhecheva, Y., Oser, C., Fleischhaker, C., Biscaldi, M., & Klein, C. (2020). Media use, sleep quality, and ADHD symptoms in a community sample and a sample of ADHD patients aged 8 to 18 years. *Journal of Attention Disorders*, *24*(4), 576–589. https://doi.org/10.1177/1087054718802014

Tong, L., Xiong, X., & Tan, H. (2016). Attention-deficit/hyperactivity disorder and lifestyle-related behaviors in children. *PLOS ONE*, *11*(9), e0163434. https://doi.org/10.1371/journal.pone.0163434

Verma, L., Agarwal, V., Arya, A., Gupta, P. K., & Mahour, P. (2022). Gaming addiction in children and adolescents with attention-deficit hyperactivity disorder and disruptive behavior disorders. *Annals of Indian Psychiatry*, *6*(2), 149. https://doi.org/10.4103/aip.aip_77_21

Vysniauske, R., Verburgh, L., Oosterlaan, J., & Molendijk, M. L. (2020). The effects of physical exercise on functional outcomes in the treatment of ADHD: A meta-analysis. *Journal of Attention Disorders*, *24*(5), 644–654. https://doi.org/10.1177/1087054715627489